계량경제학 강의 7

단순회귀모형(2)

$\boxed{\hat{\beta}_1 \quad \hat{\beta}_2}$

(순서) i) 모집단 특징 . β_1 β_2 σ^2

ii) $\hat{\beta}_1$. $\hat{\beta}_2$. $\hat{\sigma}^2$

iii) $E(\hat{\beta}_1) = \beta_1$, $E(\hat{\beta}_2) = \beta_2$.

$E(\hat{\sigma}^2) = \sigma^2$

iv) $var(\hat{\beta}_1)$, $var(\hat{\beta}_2)$. $var(\hat{\sigma}^2)$

v) $\hat{\beta} \sim N$

$\hat{\sigma}^2 \sim$

vi) 가설검정

1. OLS 추정량의 평균

X_i : non-stochastic

$\hat{\beta}_2 = \frac{\sum x_i y_i}{\sum x_i^2}$ ⇒ let $k_i \equiv \frac{x_i}{\sum x_i^2}$

$\hat{\beta}_2 = \sum k_i y_i$

$$E(\widehat{\beta_2}) = E\left(\frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2}\right)$$

$= \underline{\underline{\text{Yes.}}}\ ,\ \beta_2$

∴ $\hat{\beta}_2$은 β_2의 unbiased Estimator.

$$E(\widehat{\beta_1}) = E(\bar{Y} - \widehat{\beta_2}\bar{X})$$

$\frac{1}{n}\sum Y_i$ $\quad \frac{1}{n}\sum X_i$ $\quad$ (H.W)

$=$ $\qquad = \frac{x_i}{\sum x_i^2}$

※ Note $\quad \boxed{k_i = \frac{x_i}{\sum x_i^2}}$, $x_i = X_i - \bar{X}$

① $\sum k_i = 0$ $\quad$ ② $\sum k_i^2 = \frac{1}{\sum x_i^2}$

③ $\sum k_i X_i = 1$

$E(\hat{\beta}_2) = \beta_2$

① $\sum k_i = 0$

$$\sum k_i = \sum\left(\sum \frac{x_i}{x_i^2}\right) = \frac{\sum x_i}{\sum x_i^2} = \frac{\sum(X_i - \bar{X})}{\sum x_i^2} = \frac{\sum X_i - n\cdot\bar{X}}{\sum x_i^2} = \frac{\sum X_i - \sum X_i}{\sum x_i^2}$$

$\left(\bar{X} = \frac{1}{n}\sum X_i \quad \boxed{n\cdot\bar{X}} = \sum X_i\right)$

∴ β_1, β_2: 불편추정량(unbiased estimator)

$$\hat\beta_2 = \frac{\sum x_i y_i}{\sum x_i^2} \qquad E(\hat\beta_2) = \beta_2\ ?$$

let $k_i \equiv \dfrac{x_i}{\sum x_i^2}$

$$\Rightarrow \hat\beta_2 = \sum k_i \cdot y_i$$

$$\boxed{① \ \sum k_i = 0}$$

$$\boxed{② \ \sum k_i^2 = \frac{1}{\sum x_i^2}}$$

$$k_i = \frac{x_i}{\sum x_i^2}$$

$$\sum\left(\frac{x_i}{\sum x_i^2}\right)^2 = \frac{\cancel{\sum x_i^2}}{(\sum x_i^2)^{\cancel{2}}} = \frac{1}{\sum x_i^2}$$

$$\boxed{③ \ \sum k_i X_i = 1}$$

$$\sum k_i X_i = \frac{\sum x_i X_i}{\sum x_i^2}, \qquad \text{cf) } x_i = X_i - \bar X$$

$$= \frac{\sum x_i (x_i + \bar X)}{\sum x_i^2} = \frac{\cancel{\sum x_i^2}}{\cancel{\sum x_i^2}} + \frac{\bar X \sum x_i}{\sum x_i^2} \quad //$$

$$\sum x_i = \sum(X_i - \bar X) = \sum X_i - \sum \bar X = n\cdot\bar X - n\cdot\bar X = 0$$

$$\hat\beta_2 = \sum k_i y_i$$

$$= \sum k_i (Y_i - \bar Y) \qquad \text{pop. eqn}$$

$$= \sum k_i(\beta_1 + \beta_2 X_i + e_i) - \bar Y \sum k_i \quad //0$$

$$\hat\beta_2 = \beta_1 \sum k_i + \beta_2 \sum k_i X_i + \sum k_i e_i$$

$$\sum k_i = 0 \text{ by ①}$$

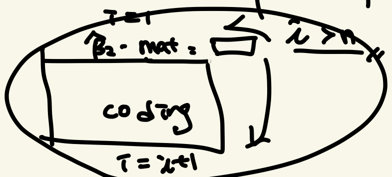
Then, $E(\cdot)$ on both side,

$$E(\hat\beta_2) = \beta_2 + E\left(\sum k_i e_i\right)$$

($\because X_i$: non-stochastic)

$$= \beta_2 + \sum k_i E(e_i) \quad //0 \text{ by assup ①}$$

$$\therefore E(\hat\beta_2) = \beta_2$$



2. OLS 추정량의 분산

by definition

$$\boxed{\hat{\beta}_2 = \beta_2 + \Sigma k_i e_i}$$

$$Var(\hat{\beta}_2) = E((\hat{\beta}_2 - \underset{=E(\hat{\beta}_2)}{\beta_2})^2) =$$

$$= E\left[(\Sigma k_i e_i)^2\right] \qquad \because E(e_i e_j) = 0$$

$$= E\left[\underbrace{k_1^2 e_1^2}_{\sigma^2} + k_2^2 e_2^2 + \dots + \underbrace{k_n^2 e_n^2}_{\sigma^2} + \cancel{2k_1 e_1 k_2 e_2 + \cdots}\right]$$

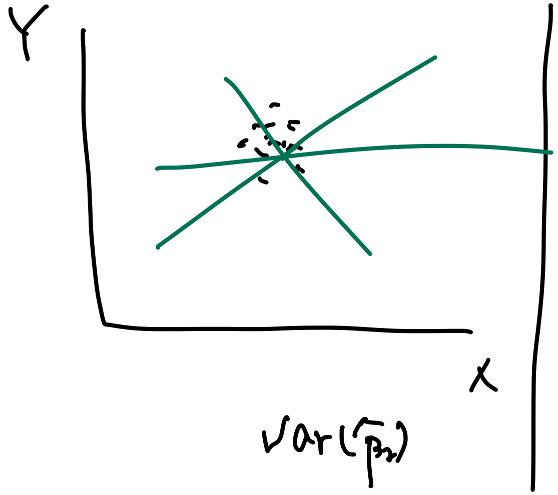
$$= k_1^2 E(e_1^2) + k_2^2 E(e_2^2) + \cdots$$

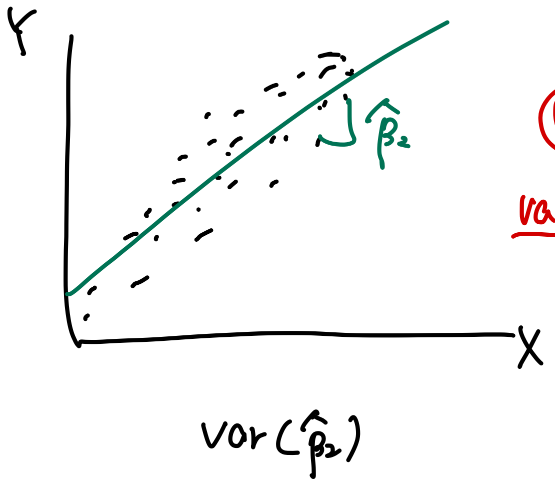
$$= k_1^2\sigma^2 + k_2^2\sigma^2 + \cdots + k_n^2\sigma^2 = \sigma^2 \Sigma k_i^2 \quad \text{by (ii)}$$

$$= \frac{\sigma^2}{\Sigma x_i^2}$$

$$\hat{\beta}_2 = \frac{\widehat{cov}(X,Y)}{\widehat{var}(X)}$$

$$Var(\hat{\beta}_2) = \boxed{\frac{\sigma^2/(n-1)}{\sum x_i^2/(n-1)}} = \frac{\sigma^2/(n-1)}{\widehat{Var}(X)}$$





$n \uparrow$

$var(x) \uparrow$

3. OLS 추정량의 분포

$(e_i | X_i \sim N)$, $(Y_i | X)$

$e_i \sim iid N(0, \sigma^2)$이라고 가정해보자

$\rightarrow \underbrace{Y_i}_{} \overset{= \beta_1 + \beta_2 X_i + e_i}{\sim} iid N(\underbrace{\beta_1 + \beta_2 X_i}_{E(Y_i)}, \underbrace{\sigma^2}_{Var(Y_i)})$

$\rightarrow \widehat{\beta_2}$도 확률변수 $\mathrm{Y_i}$의 함수이므로 정규분포를 따른다.

$= \frac{\sum x_i Y_i}{\sum x_i^2}$ → 분포

$$\boxed{\widehat{\beta_2} \sim N(\beta_2, \frac{\sigma^2}{\sum x_i^2})}$$

//

$\hat{\sigma}^2$?　　$Var(\hat{\sigma}^2)$?　　분포?

3. OLS 추정량의 분포

$Y_i = \beta_1 + \beta_2 X_i + e_i$

cf) $Y_i = \mu + e_i$

$\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{\boxed{n-1}}$

모분산이 알려져 있지 않은 경우 표본분산을 대신 사용한다.

(예전) $\boxed{Y_i = \mu + e_i}$ $e_i \sim iid\, N(0, \sigma^2)$

$\frac{\sum \hat{e}_i^2}{\sigma^2} \sim \chi^2(n-1)$

$$\widehat{\sigma_2} = \frac{\sum_{i=1}^{n} \hat{e}_i^{\,2}}{(n-2)} = \frac{\sum_{i=1}^{n} (Y_i - \widehat{\beta_1} - \widehat{\beta_2} X_i)^2}{(n-2)} \to \frac{\sum \hat{e}_i^2}{\sigma^2} \sim \chi^2(n-2)$$

$* \; E(\hat{\sigma}^2) = \sigma^2$

$var(\hat{\sigma}^2) = ?$

$$\widehat{\beta_2} \sim N\left(\beta_2, \frac{\sigma^2}{\sum x_i^2}\right) \to$$

$\frac{(n-2)\hat{\sigma}^2}{\sigma^2} \sim \chi^2(n-2)$

$E\left(\frac{(n-2)\hat{\sigma}^2}{\sigma^2}\right) = n-2$ $\quad E(\hat{\sigma}^2) = \sigma^2$

따라서,

$var\left(\frac{(n-2)\hat{\sigma}^2}{\sigma^2}\right) = 2 \cdot (n-2)$

$\frac{(n-2)^2}{\sigma^4} var(\hat{\sigma}^2) = 2 \cdot (n-2)$

$var(\hat{\sigma}^2) = \frac{2 \cdot \sigma^4}{n-2}$

4. 가설검정

예) $\boxed{Y_i = \beta_1 + \beta_2 X_i + e_i}$

Y_i: 식품에 대한 지출, X_i: 총지출, n =200

$\boxed{1}$ H_0: $\beta_2 = 1$ vs H_1: $\beta_2 < 1$

$\boxed{2}$ If $\beta_2 \ll 1$, we reject H_0

$\alpha = 5\%$ 유의수준

검정통계량) H_0 下,

① 계산가능

② 분포를 알아야 한다.

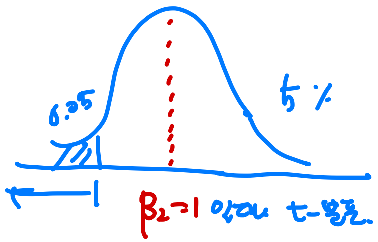
모른다면?

$$\hat{\beta}_2 \sim N\left(\beta_2, \frac{\sigma^2}{\sum x_i^2}\right)$$

t-value

$$= \frac{\hat{\beta}_2 - \beta_2}{\sqrt{\frac{\hat{\sigma}^2}{\sum x_i^2}}} \sim t(n-2)$$

〃 1



4. 가설검정

예) $Y_i = \beta_1 + \beta_2 X_i + e_i$

Y_i: 식품에 대한 지출, X_i: 총지출, n =200

H_0: $\beta_2 = 0$ vs H_1: $\beta_2 \neq 0$

$\boxed{1}$

$\boxed{2}$ $\alpha = 0.05$

$$z = \frac{\hat{\beta}_2 - \beta_2}{\sqrt{\frac{\sigma^2}{\sum x^2}}} \sim N(0,1)$$

1/0